

Tensor Analysis

7. Matrices

A matrix A of order m by n is an array of quantities a_{pp} called elements, which are arranged in m rows and n columns and generally denoted by

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \text{ or } \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

or in abbreviated form by $[a_{pq}]$ or (a_{pq}) , $p = 1, \dots, m$; $q = 1, \dots, n$. We use the former notation, $[a_{pq}]$, unless otherwise stated or implied. If $m = n$, the matrix is a square matrix of order n or simply order m . If $m = 1$, it is a row matrix or row vector; if $n = 1$, it is a column matrix or column vector.

The diagonal of a square matrix containing the elements $a_{11}, a_{22}, \dots, a_{nn}$ is called the principal or main diagonal. A square matrix whose elements are equal to one in the principal diagonal and zero elsewhere is called a unit matrix and is denoted by I . A null matrix, denoted by O , is a matrix whose elements are all zero.

Matrix Algebra

Suppose $A = [a_{pq}]$ and $B = [b_{pq}]$ are matrices having the same order (m by n). Then the following definitions apply:

(1) $A = B$ if $a_{pq} = b_{pq}$ for all p and q .

(2) The sum S and difference D of A and B are the matrices defined by

$$S = A + B = [a_{pq} + b_{pq}], \quad D = A - B = [a_{pq} - b_{pq}]$$

That is, the sum $S = A + B$ [difference $D = A - B$] is obtained by adding (subtracting) the corresponding elements of A and B . element of A is multiplied by A . The transpose of a matrix A is a matrix A^T , which is formed from A by interchanging its rows and columns. Thus, if $A = [a_{pq}]$, then $A^T = [a_{qp}]$.

Matrix Multiplication

Now suppose A and B are two matrices such that the number of columns of A is equal to the number of rows of B . say A is an $m \times p$ matrix and B is a $p \times n$ matrix. Then the product of A and B is defined and the product, denoted by AB , is the matrix whose i -entry is obtained by multiplying the elements of row i of A by the corresponding elements of column j of B and then adding. Thus, if $A = [a_i]$ and $B = [b_j]$. then $AB = [c_{ij}]$ where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ip}b_{pj} = \sum_{k=1}^p a_{ik}b_{kj}$$

Matrices whose product is defined are called conformable.

Determinants

Consider an n -square matrix $A = [a_{ij}]$. The determinant of A is denoted by $|A|$, $\det A$, $|a_v|$, or $\det [a_{ij}]$. The reason may be familiar with the definition of $\det A$ when $n \leq 3$. The general definition of $\det A$ follows:

$$|A| = \sum_{\sigma \in S_n} (\text{sgn } \sigma) a_{1\sigma(1)} a_{2\sigma(2)} \dots a_{n\sigma(n)}$$

Here S_R consists of all permutations σ of $\{1, 2, \dots, n\}$, and sign $\sigma = \pm 1$ according as σ is an even or odd permutation.

One main property of the determinant follows:

Proposition:

Let $P = AB$ where A and B are n -square matrices, Then

$$\det P = (\det A)(\det B)$$

Inverses:

The inverse of a square matrix A is a matrix, denoted by A^{-1} , such that $AA^{-1} = A^{-1}A = I$

where I is the unit matrix. A necessary and sufficient condition that A^{-1} exists is that $\det A \neq 0$.

If $\det A = 0$, then A is called singular, otherwise A is called nonsingular.